

Aflevering 4

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Exercise 7: (Exercise 1 from Exam of May 2023) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the 2π -periodic function that for $-\pi \leq x < \pi$ is given by

$$f(x) := \cos\left(\frac{x}{2}\right).$$

1. Specify formulas for calculating the coefficients a_n and b_n of the corresponding real Fourier series

$$F(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

of the function f which explain that for all $n \geq 1$ we have that $b_n = 0$ and show that

$$a_n = \frac{(-1)^n}{\pi(\frac{1}{4} - n^2)} \quad \text{for each } n \geq 1.$$

Find also a_0 .

1 Since $f(x)$ is an even function, we can use Theorem 1.8 (with $a = \pi$), to find the coefficients:

$$b_n = 0$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} \cos\left(\frac{x}{2}\right) dx = \frac{2}{\pi} \left[\sin\left(\frac{x}{2}\right) \right]_0^{\pi} = \frac{2}{\pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} \cos\left(\frac{x}{2}\right) \cos(nx) dx, \quad \boxed{\cos(a) \cos(b) = \frac{1}{2} (\cos(a-b) + \cos(a+b))}$$
$$= \frac{1}{\pi} \left(\int_0^{\pi} \cos\left(\frac{x}{2} - nx\right) dx + \int_0^{\pi} \cos\left(\frac{x}{2} + nx\right) dx \right)$$

$$= \frac{1}{\pi} \left(\left(\frac{1}{\frac{1}{2} - n} \right)' \left[\sin\left(\frac{x}{2} - nx\right) \right]_0^{\pi} + \left(\frac{1}{\frac{1}{2} + n} \right)' \left[\sin\left(\frac{x}{2} + nx\right) \right]_0^{\pi} \right)$$

$$= \frac{1}{\pi} \left(\frac{-2}{2n-1} \cdot \sin\left(\frac{\pi}{2} - n\pi\right) + \frac{2}{2n+1} \sin\left(\frac{\pi}{2} + n\pi\right) \right),$$

$$\text{" } \cos(n\pi) = (-1)^n \text{ "}$$

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$$= \frac{(-1)^n}{\pi} \left(\frac{2}{2n+1} - \frac{2}{2n-1} \right) = \frac{(-1)^n}{\pi \left(\frac{1}{4} - n^2 \right)}$$

2. For which values of $x \in [-\pi, \pi]$ is the real Fourier series F convergent, and for which values $x \in [-\pi, \pi]$ does $F(x) = f(x)$? If $F(x) \neq f(x)$, what is the value of $F(x)$? Is the real Fourier series F uniformly convergent to f on the interval $[-\pi, \pi]$? Does the real Fourier series F converge to f in the norm of $L^2([-\pi, \pi])$?

From Theorem 1.22 we know $F(x)$ converges to $f(x)$ in x if:

- f is continuous
- f is 2π -periodic
- f is differentiable in x

$\cos\left(\frac{x}{2}\right)$ satisfies these conditions for $x \in [-\pi, \pi]$

$$\text{So } F(x) = f(x)$$

From Theorem 1.30 we know

$F(x)$ converges uniformly

if $f(x)$ is also piecewise smooth.

Since $f(x)$ is continuous and differentiable

it is also piecewise smooth, so

$F(x)$ converges uniformly.

From Theorem 1.35 we know

$F(x)$ converges to f in the

norm if f is in $L_2([-\pi, \pi])$

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$[-\pi, \pi]$

$$\int_{-\pi}^{\pi} |f(x)|^2 dx < \infty$$

$$\int_{-\pi}^{\pi} \cos^2\left(\frac{x}{2}\right) dx, \quad \cos^2(u) = \frac{1}{2}(\cos(2u) + 1)$$

$$= \frac{2}{2} \int_0^{\pi} \cos(x) + 1 dx$$

$$= [\sin(x)]_0^{\pi} + [x]_0^{\pi} = \pi < \infty$$

So $F(x)$ converges to f in the norm.

3. Find the value of the infinite sum

$$\sum_{n=1}^{\infty} \frac{1}{(n - \frac{1}{2})^2 (n + \frac{1}{2})^2}$$

To find the value we use
Theorem 1.39

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx = 2|a_0|^2 + \sum_{n=1}^{\infty} (|a_n|^2 + |b_n|^2)$$

$$\Rightarrow 1 = 2 \left| \frac{2}{\pi} \right|^2 + \sum_{n=1}^{\infty} \left| \frac{(-1)^n}{\pi \left(\frac{1}{4} - n^2 \right)} \right|^2$$

We start by calculating the parentheses:

$$\left(n - \frac{1}{2}\right)^2 \left(n + \frac{1}{2}\right)^2$$

$$= \left(n^2 + \frac{1}{4} - n\right) \left(n^2 + \frac{1}{4} + n\right)$$

$$= n^4 + \frac{n^2}{4} + n^3 + \frac{n^2}{4} + \frac{1}{16} + \frac{n}{4} - n^3 - \frac{n}{4} - n^2$$

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$$= n^4 - \frac{n^2}{2} + \frac{1}{16}$$

$$\left(\frac{1}{4} - n^2\right)^2 = \frac{1}{16} + n^4 - \frac{n^2}{2} = \left(n - \frac{1}{2}\right)^2 \left(n + \frac{1}{2}\right)^2$$

$$(-1)^{2n} = 1 \quad \text{always even}$$

So we get:

$$\Rightarrow 1 = \frac{8}{\pi^2} + \frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{\left(n - \frac{1}{2}\right)^2 \left(n + \frac{1}{2}\right)^2}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{\left(n - \frac{1}{2}\right)^2 \left(n + \frac{1}{2}\right)^2} = \pi^2 - 8$$

Exercise 8: (Exercise 2 from Exam of May 2023) Let $f: \mathbb{R} \rightarrow \mathbb{C}$ be the 2-periodic function that for $-1 \leq x < 1$ is given by

$$f(x) = \begin{cases} 1 & \text{for } -1 \leq x < 0 \\ -1 & \text{for } 0 \leq x < 1. \end{cases}$$

1. By integrating, calculate the coefficients c_n of the corresponding complex Fourier series

$$F(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x}$$

of the function f .

We use Theorem 1.18 to calculate c_n .

$$\begin{aligned} c_n &= \frac{1}{2} \left(\int_{-1}^0 e^{-in\pi t} dt - \int_0^1 e^{-in\pi t} dt \right) \\ &= \frac{1}{2} \left(\frac{1}{in\pi} e^0 + \frac{1}{in\pi} e^{in\pi} + \frac{1}{in\pi} e^{-in\pi} - \frac{1}{in\pi} e^0 \right) \\ &= \frac{1}{2in\pi} \left(-2 + \underbrace{e^{in\pi} + e^{-in\pi}}_{2 \cos(n\pi)} \right) \\ &= \frac{(-1)^n - 1}{in\pi} \end{aligned}$$

$$= \frac{(-1)^n - 1}{in\pi}$$

$$2 \cos 6\pi = 2(-1)$$

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$$F(x) = \sum_{n=-\infty}^{\infty} \frac{(-1)^n - 1}{in\pi} e^{in\pi x}$$

2. Determine whether the complex Fourier series F of f converges uniformly on $\mathbb{R}$ towards f , converges pointwise towards f , and if F converges towards f with respect to the norm of $L^2([-1, 1])$. Justify the answers.

From Theorem 1.30 we know F doesn't converge to f uniformly since f is not continuous.

If we extend $f(x)$ to be 2π -periodic, we can use Theorem 1.28 (B) to see that $F(x)$ converges for all x , if f is of bounded variation, which it is, since f is piecewise smooth.

F converges to f in the norm if f is in L^2 .

$$\int_{-1}^0 \|1\|^2 dt + \int_0^1 \|1\|^2 dt = 0 - (-1) + 1 - 0 = 2 < \infty$$

So F converges to f in the norm.

3. The vector space $L^2([-1, 1])$ is endowed with the usual inner product

$$\langle g, h \rangle := \int_{-1}^1 g(x) \overline{h(x)} dx \quad \text{for any } g, h \in L^2([-1, 1]).$$

Determine the function $g(x)$ in the vector space

$$V := \text{span}\{e^{-2i\pi x}, e^{-i\pi x}, 1, e^{i\pi x}, e^{2i\pi x}\}$$

that is at the minimum distance possible from the above function $f \in L^2([-1, 1])$. Also check whether $f(x)$ is orthogonal on the function $h(x) = e^{2i\pi x} \in L^2([-1, 1])$.

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We can use the vector space to make the Fourier series:

$$g(x) = \sum_{n=-2}^2 c_n e^{in\pi x}$$

$$= 0 \cdot e^{-2i\pi x} + \frac{-2}{-i\pi} e^{-i\pi x} + 0 \cdot e^0 + \frac{-2}{i\pi} e^{i\pi x}$$

$$= \frac{2}{i\pi} (e^{-i\pi x} - e^{i\pi x})$$

$$= \frac{2}{i\pi} \cdot -2i \sin(\pi x)$$

$$= \frac{-4}{\pi} \sin(\pi x)$$

To check if f is orthogonal to h
we use

$$0 = \langle f, h \rangle = c_2 = \frac{(-1)^2 - 1}{2i\pi} = 0 \quad \checkmark$$

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